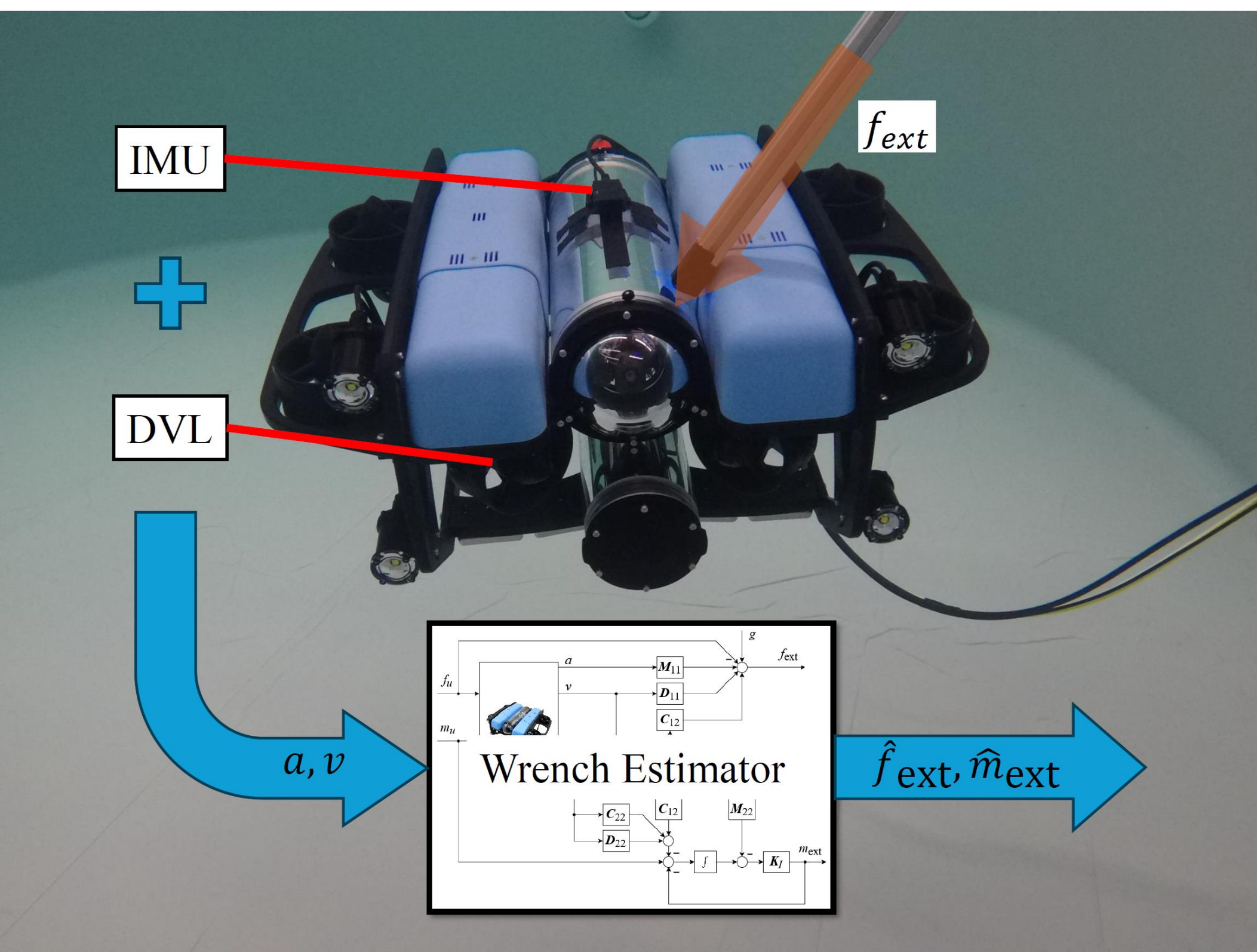


Towards Dexterous Underwater Manipulation with Small-scale Robots: External Wrench Estimation and Interaction Detection

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Motivation

- Autonomous underwater manipulation
 - Interaction forces and torques retroact on the robot
- Crucial for planning and control

But: Underwater force/torque-sensors are expensive
for *small-scale* systems:
sensor cost > vehicle cost

Goal

Can we estimate the external wrench acting on an underwater robot with *onboard motion* sensors?

Key Challenges

- Incomplete and noisy measurements
- Nonlinear system dynamics

Concept

Hybrid **external wrench estimation** from aerial drones:

- *Acceleration-based* estimation for forces

$$\hat{f}_{\text{ext}} = M_{11}\dot{v} + C_{12}(v)\omega + D_{11}(v)v + g_v(\eta) - f_u$$

- *Momentum-based* estimation for torques

$$\hat{m}_{\text{ext}} = K_I \left[M_{22}\omega - \int_0^t (m_u - N_m - \hat{m}_{\text{ext}}) ds \right]$$

with

$$N_m = -C_{12}(v)v + C_{22}(\omega)\omega + D_{22}(\omega)\omega.$$

Interaction detection using Gaussian mixture model

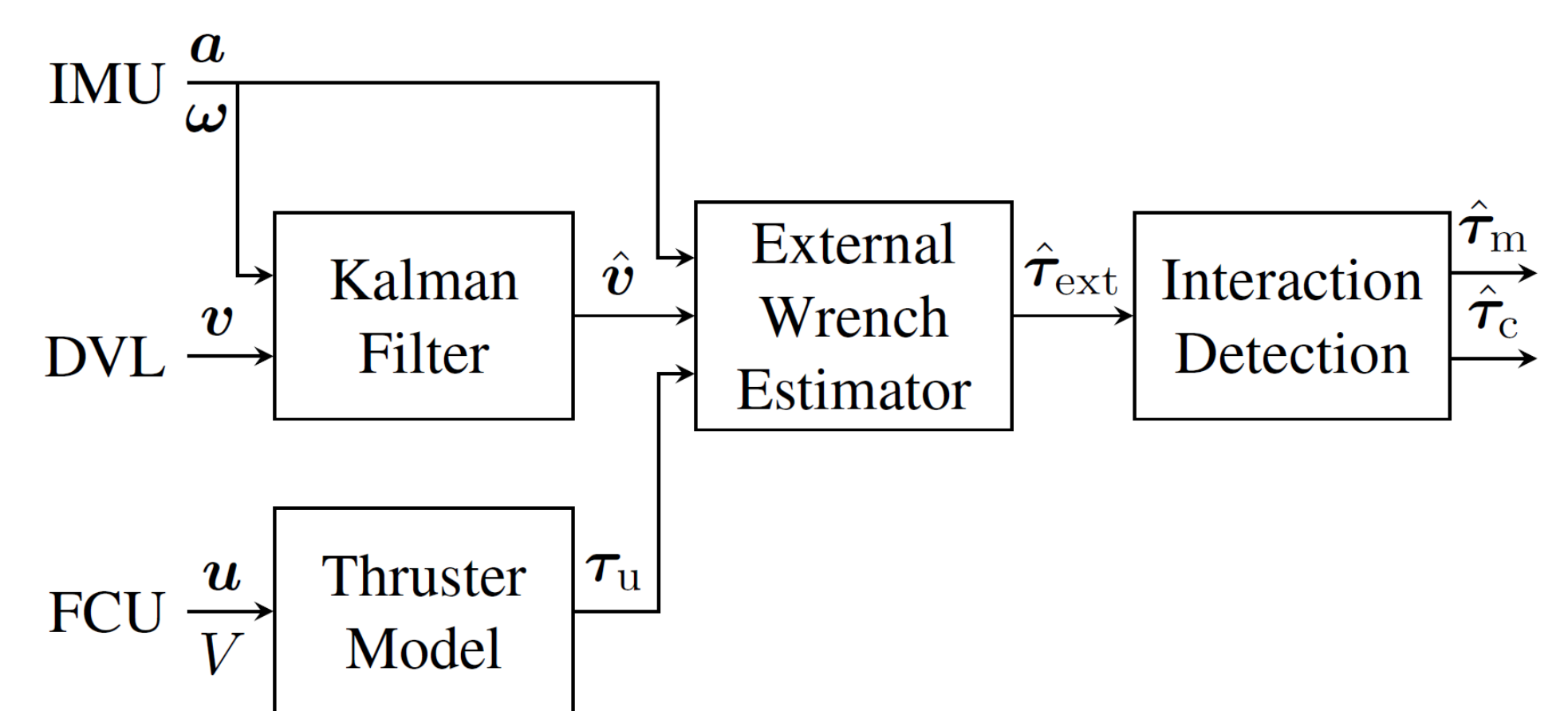
Assumption:

External wrench =

Quasi-static background from slow changing water currents

+

dynamic foreground from robot interaction



The whole pipeline consists of the core elements external wrench estimator and interaction detection.

Experimental Analysis

4.5m x 4.5m x 1.2m fresh water tank

Lifting payload weight under changing current disturbances

→ 3 different payload weights

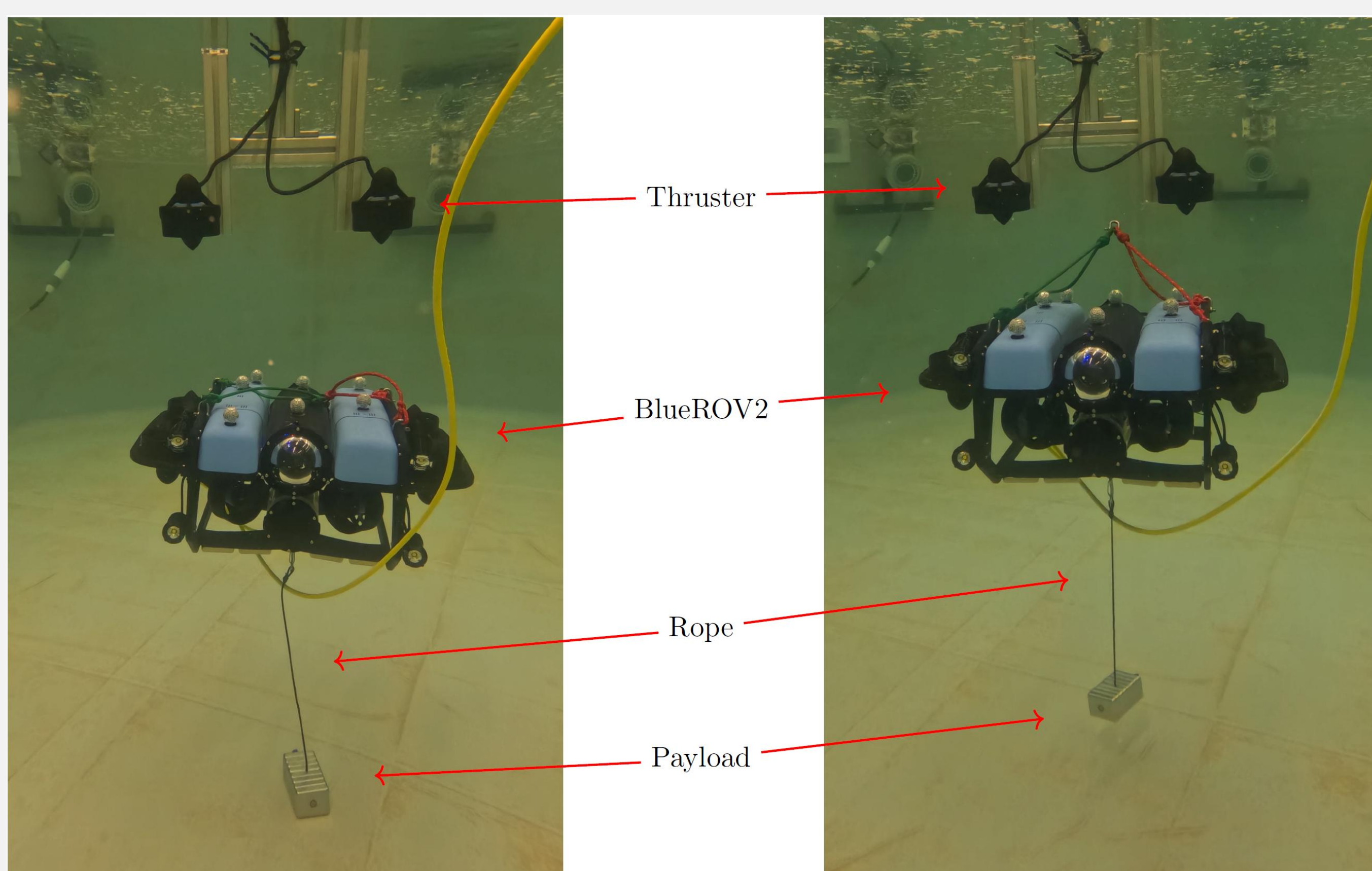
→ 4 different current strengths

Same parameters for all experimental settings

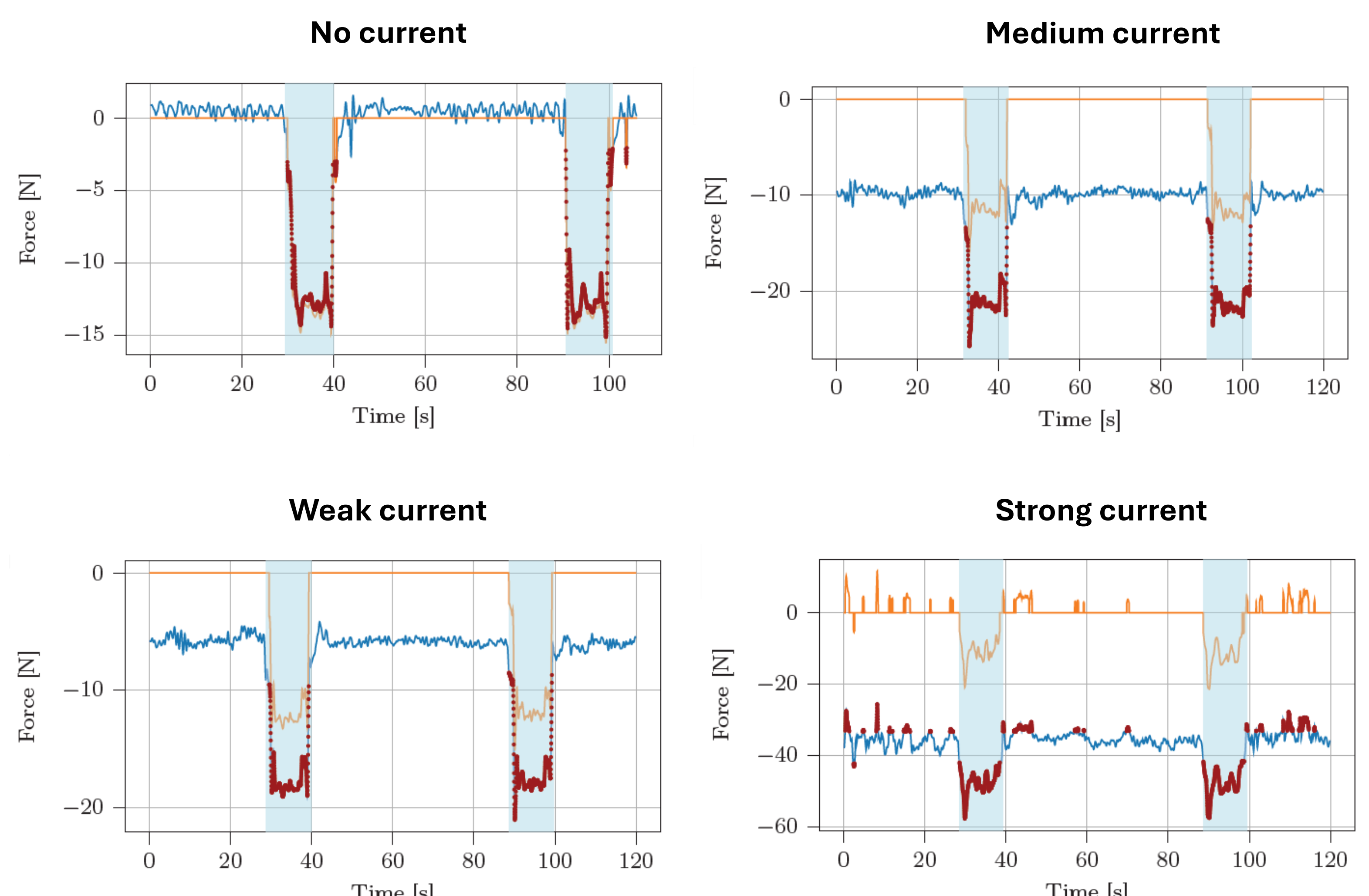
Results

- Good detection for medium and high payload...
- No performance loss with no, weak, and medium currents
- Low performance for low payload and strong currents

Current	Metric	Payload		
		6 Weights	4 Weights	2 Weights
No	TPR	95.1 %	95.2 %	49.9 %
	FPR	1.1 %	1.4 %	0.9 %
Weak	TPR	91.3 %	89.5 %	64.6 %
	FPR	0.0 %	0.5 %	5.2 %
Medium	TPR	94.7 %	95.3 %	63.0 %
	FPR	0.0 %	0.2 %	0.1 %
Strong	TPR	93.5 %	96.8 %	72.4 %
	FPR	16.4 %	3.9 %	11.3 %



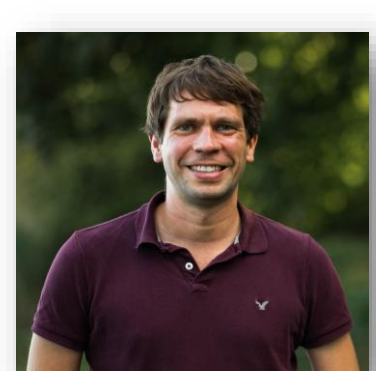
BlueROV2 lifting payload under influence of current disturbance from thrusters. In lower setpoint (left) payload rests on ground. In upper setpoint (right) payload is carried by BlueROV2.



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